

Impact of Trading Activity on intraday Volatility and Liquidity

Victoria Godio, Baptiste Messant & Raphaël Perilhou

Toulouse School of Economics

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1 Discovery

1.1 Context and Motivation

According to Black and Scholes (1973), volatility is a very important variable in option pricing and more generaly in finance. Additionaly, Markowitz (1952) also highlighted the importance of volatility in portfolio optimisation, making a link between volatility risk management and diversification. Looking at those research we can see that volatility of stocks play a central role in finance and having the best estimation of it can be crucial. This is why in this paper we choose to study if computing the variance with high frequency data can improve the quality of its forecast compared to standard methods.

For decades, the major approach for volatility modeling was based on GARCH(1,1) models introduced by Bollerslev (1986). In this method the variance is computed as a function of past squared returns and past conditional variances using only one daily observation per trading day. Even though this method is very useful it suffers from a limitation : by using only one observation per day the daily squared returns become noisy for estimating the true underlying volatility. As Andersen and Bollerslev (1998) demonstrated, this noise is largely responsible for the apparent poor out-of-sample forecasting performance of GARCH models when evaluated against daily squared returns.

Since the late 1990s, the rapid expansion of electronic trading has made high-frequency financial data widely available. These data record transactions and quotes at the second or even millisecond level. This development has opened up new possibilities for measuring and forecasting volatility. Using high frequency data and summing squared intraday returns it is possible to compute the realized volatility. Additionally, Andersen et al. (2003) demonstrated that forecasts produced using simple time-series models applied to realized volatility substantially outperform those produced using traditional daily GARCH-type models. Their work established realized volatility as a superior basis for volatility forecasting.

1.2 The Problem of Microstructure Noise

A practical complication arises when working with very high-frequency data. At the finest sampling frequencies, observed prices are contaminated by market microstructure noise. Hansen and Lunde (2006) provided extensive empirical evidence that this noise is both time-dependent and correlated with the efficient price process. As demonstrated by Zhang et al. (2005), in the presence of such noise, the standard realized volatility estimator becomes inconsistent. It means its bias increases linearly with the number of intraday observations. This creates a fundamental trade-off between bias and variance when choosing the sampling frequency. The literature therefore recommends a moderate frequency

of 5 minutes as an effective compromise (Hansen and Lunde, 2006; Barndorff-Nielsen and Shephard, 2002).

1.3 The Heterogeneous Market Hypothesis and the HAR Model

An important insight into the dynamics of realized volatility comes from the Heterogeneous Market Hypothesis (HMH) proposed by Müller et al. (1997). This hypothesis posits that financial markets are populated by agents operating at fundamentally different time horizons from intraday traders and market makers to weekly portfolio rebalancers and monthly institutional investors. Each group reacts to information differently and over different time scales, generating a cascade of volatility from long to short horizons. Empirical evidence shows that long-horizon volatility exerts a stronger causal influence on short-horizon volatility than the reverse (Müller et al., 1997; Lynch and Zumbach, 2003).

Motivated by the HMH, Corsi (2009) proposed the Heterogeneous Autoregressive model of Realized Variance (HAR-RV). This model expresses future realized variance as a linear function of three components: the daily, weekly (5-day average), and monthly (22-day average) realized variance[1]. Despite its remarkably simple structure it is essentially a constrained AR(22) model estimated by ordinary least squares. The HAR-RV successfully reproduces the key empirical features of variance dynamics, including long memory, fat tails, and self-similarity. Moreover, numerous empirical studies have confirmed its strong out-of-sample forecasting performance across different asset classes and markets (Corsi, 2009; Andersen et al., 2007).

The HAR-RV has since become a standard benchmark in the realized volatility forecasting literature (Bollerslev et al., 2016), and various extensions have been proposed, including the incorporation of jump components (Andersen et al., 2007), asymmetric effects, and additional explanatory variables.

1.4 Research Question and Hypotheses

In this project, we study the following question: **Can high frequency data and market activity variables improve the forecasting of S&P500 volatility compared to the classical GARCH approach ?**

Our hypotheses are the following:

H1 : The HAR-RV model produces superior out of sample forecasts compared to a standard GARCH(1,1) model estimated on daily returns.

[1] A simple explanation can be found here.

H2 : Augmenting the HAR-RV model with trading volume and the number of transactions as additional regressors improves forecasting accuracy.

2 Data preparation

2.1 Data Sample

For our project we choose to work with data from a US index, the S&P500. In order to have consistent data at the low frequency we use a free API from the "massive.com" website. This API allows us to download data of an S&P500 ETF (code : SPY) at the one minute frequency from April 8, 2024 to to July 10, 2024. The date range we got is constrained by the limitations of the free version of the API. Although this restricts the statistical power of our analysis, the pipeline we build can be applied to a longer dataset to obtain more informative results. Each observation contains eight variables: the timestamp, the open, high, low, and close prices, the trading volume (number of shares traded), the volume-weighted average price (VWAP), and the number of transactions executed within the 1-minute interval.

2.2 Modifications

In order to work with consistent data we have done some modification in the dataset.

- Clean the data : We remove the potential NA and duplicate in the data.
- Adapt the time slot : Since we work with a world traded index we choose to work only with the time slot of the NYSE, we choose to do this since it is the time slot with most of the trades and the less gaps in the data.

At this point our dataset is composed of an S&P500 ETF data from the 4 August to 11 July 2024 between 9:30AM to 4:00PM (oppening hours of the NYSE).

3 Models of interest

3.1 GARCH(1,1)

The first model we consider is the GARCH(1,1), which serves as our classical benchmark. This model specifies the conditional variance as a function of the previous day's squared return (the "shock") and the previous day's conditional variance:

$$\sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2 \tag{1}$$

σ_t^2 is the conditional variance, and ω, α, β are parameters estimated by maximum likelihood under the assumption of normally distributed innovations.

3.2 HAR-RV

Our main model is the Heterogeneous Autoregressive model of Realized variance (HAR-RV) proposed by Corsi (2009). Unlike the GARCH, this model works with realized variance, a directly observable measure of daily variance constructed from high-frequency intraday data. Specifically, the daily realized variance for day t is defined as the sum of squared intraday log returns sampled at the 5 minutes frequency:

$$RV_t = \sum_{j=1}^{M} r_{t,j}^2 \tag{2}$$

where $r_{t,j} = \ln(P_{t,j}) - \ln(P_{t,j-1})$ are the 5 minutes log returns within day t, and $M \approx 78$ is the number of 5 minutes intervals in a standard 6.5 hours trading day.

The HAR-RV model expresses the next day's realized variance as a linear function of three variance components, each corresponding to a different investment horizon:

$$RV_t = \beta_0 + \beta_1 RV_{t-1}^{(d)} + \beta_2 RV_{t-1}^{(w)} + \beta_3 RV_{t-1}^{(m)} + \varepsilon_t \tag{3}$$

where:

- $RV_{t-1}^{(d)} = RV_{t-1}$ is the realized variance of the previous day, capturing short-term dynamics.
- $RV_{t-1}^{(w)} = \frac{1}{5}\sum_{i=0}^{4} RV_{t-1-i}$ is the average realized variance over the past 5 trading days, capturing medium-term persistence.
- $RV_{t-1}^{(m)} = \frac{1}{22}\sum_{i=0}^{21} RV_{t-1-i}$ is the average realized variance over the past 22 trading days, capturing the long-term variance trend.

3.3 Extended Model : HAR-RV-X

To investigate whether market activity carries additional predictive information for future volatility beyond the autoregressive components of realized volatility, we extend the HAR-RV model by including two measures of daily trading activity:

$$RV_t = \beta_0 + \beta_1 RV_{t-1}^{(d)} + \beta_2 RV_{t-1}^{(w)} + \beta_3 RV_{t-1}^{(m)} + \beta_4 Vol_{t-1} + \beta_5 Txn_{t-1} + \varepsilon_t \tag{4}$$

The first additional variable, Vol_{t-1}, is the total trading volume on day $t-1$, computed by summing all 1-minute volumes within the trading day. The inclusion of volume is motivated by the Mixture of Distributions Hypothesis (MDH) of Clark (1973), which posits that both volatility and volume are jointly driven by a common latent information arrival process. Days with unusually heavy trading activity signal elevated information flow, which tends to persist and translate into higher volatility on the following day.

The second variable, Txn_{t-1}, is the total number of individual transactions executed on day $t-1$, obtained by summing the 1-minute transaction counts. While volume measures the total quantity of shares exchanged, the number of transactions captures the intensity of trading activity regardless of trade size.

4 Model Building / Estimation

4.1 Frequency selection

In order to build our models, we first need to define the variables that we are going to use. To begin, we decided to compute the realised variance by aggregating our data at different time frequencies, in order to compare them on a signature plot and see if our hypothesis that computing the realised variance by aggregating data at a 5 minute frequency is valid.

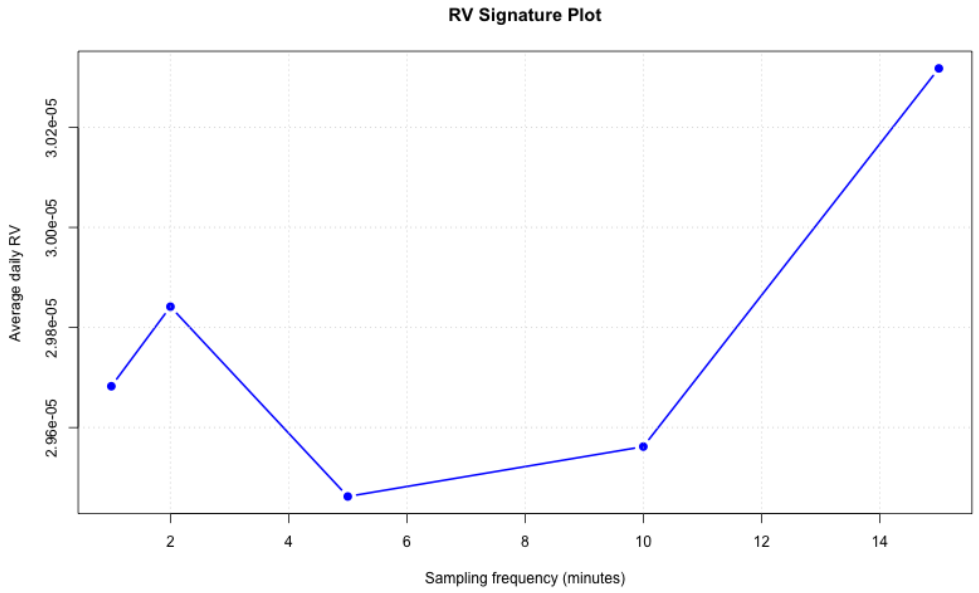


Figure 1: Variance Signature Plot

Looking at Figure 1 we can see that the signature plot is relatively stable between the 5-minute and 10-minute frequencies. This suggests that using the 5-minute frequency to compute the daily realised variance strikes a good balance between microstructure noise and information.

4.2 Variable Construction

Given that we choose to work with 5-minute frequency data, we can plot the daily realized variance over time and compare it to the classical daily variance.

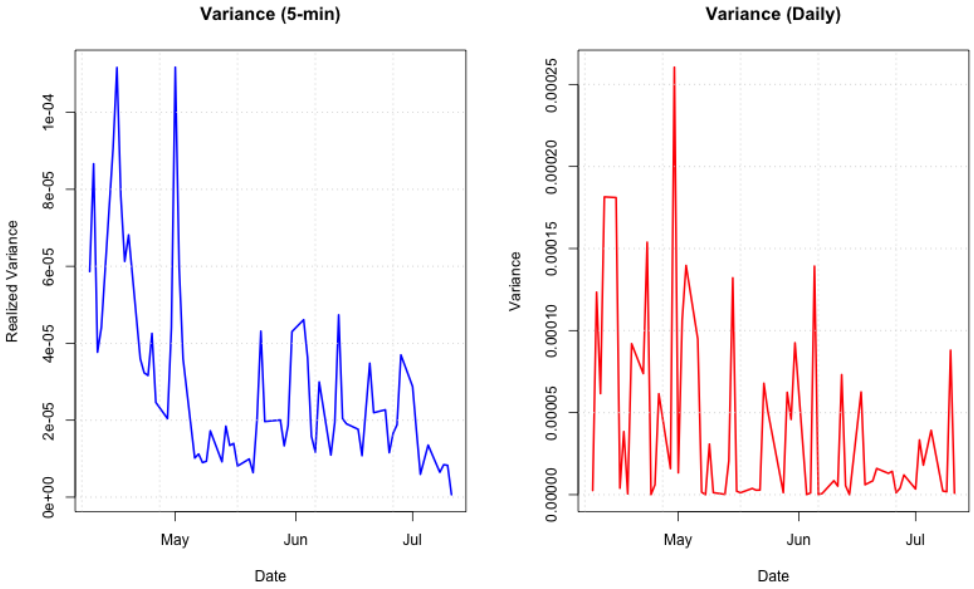


Figure 2: Variance Comparison Plot

We observe that the variance computed with the 5-minute frequency data is more stable than the classical one using daily data. Once the sampling frequency is selected, we construct the remaining variables required for estimation. The daily realized variance RV_t is computed by summing the squared 5-minute log returns within each trading day, as defined in equation (2). We then construct the weekly and monthly components of the HAR model by computing rolling averages over the past 5 and 22 trading days, respectively. Finally, we aggregate the two activity variables by summing the 1-minute volume and transaction counts within each trading day, yielding a daily total volume Vol_t and a daily transaction count Txn_t.

4.3 Descriptive Statistics

Table 1 reports summary statistics for the daily realized variance series. The mean daily RV is 2.95×10^{-5}, with a standard deviation of 2.47×10^{-5}. As shown in Figure 4, the distribution is right-skewed with a fat right tail (skewness = 1.66, excess kurtosis = 2.52), consistent with the stylized facts of realized variance series. Figure 3 shows that both RV and log(RV) display significant autocorrelations up to approximately 5 lags. This suggests short-term persistence in the variance process and supportes the daily and weekly components of the HAR specification. However, we found less support for the monthly component in the autocorrelation structure of our sample.

Table 1: Descriptive Statistics of Daily Realized Variance

Statistic	Value
Mean	2.95×10^{-5}
Std. Dev	2.47×10^{-5}
Skewness	1.66
Kurtosis	5.52
Excess Kurtosis	2.52

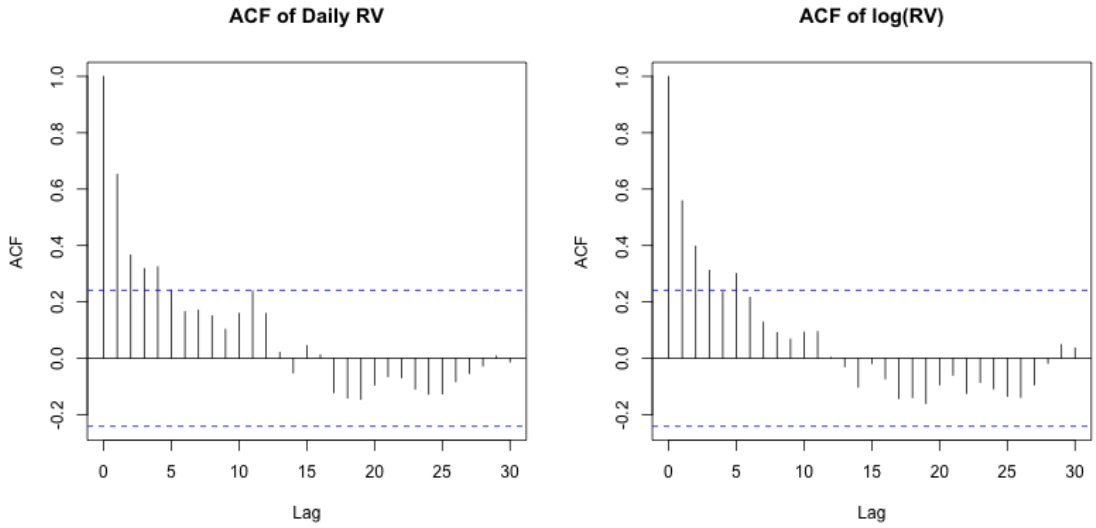


Figure 3: ACF of Daily RV (left) and log(RV) (right)

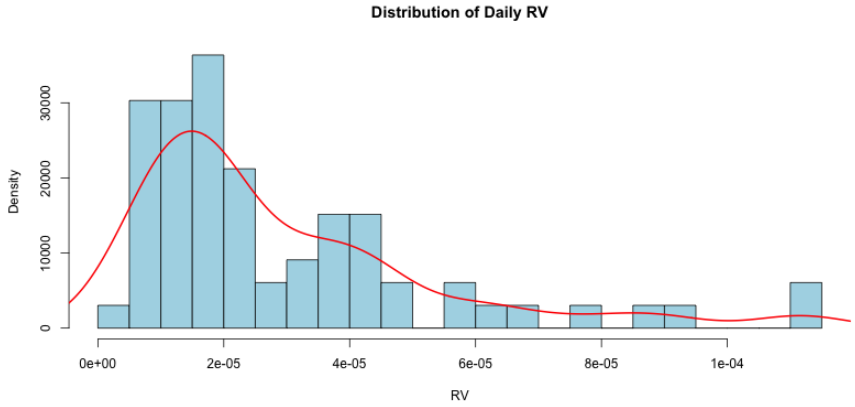


Figure 4: Distribution of Daily RV

Table 2 shows the descriptive statistics for the two daily trading activity variables. Daily volume averages around 44 million shares traded, with substantial variation (sd = 16 million). The average number of transactions per day is approximately 427,000, and also has substantial variation (sd = 141,000). The large gap between minimum and maximum values for both variables confirms that trading activity varies significantly across

days in our sample. Therefore, it motivates their inclusion as potential predictors of future volatility.

Statistic	Volume	Transactions
Mean	44248159	427098
Std. Dev	16208837	141393
Min	2752664	28976
Max	84721198	815035

Table 2: Descriptive Statistics: Daily Volume and Transactions

5 Results

5.1 In-Sample Fit

Figure 5 displays the actual daily RV against the HAR-RV fitted values over the in-sample period. The model captures the general level and gradual movements of realized variance reasonably well, but we clearly see that it smooths out the larger volatility spikes. If we look at Equation 3, we can see the fitted values as weighted averages of past RV components and therefore cannot reproduce sudden large deviations. This also explains the moderate adjusted R^2 of 0.115 reported in Table 3.

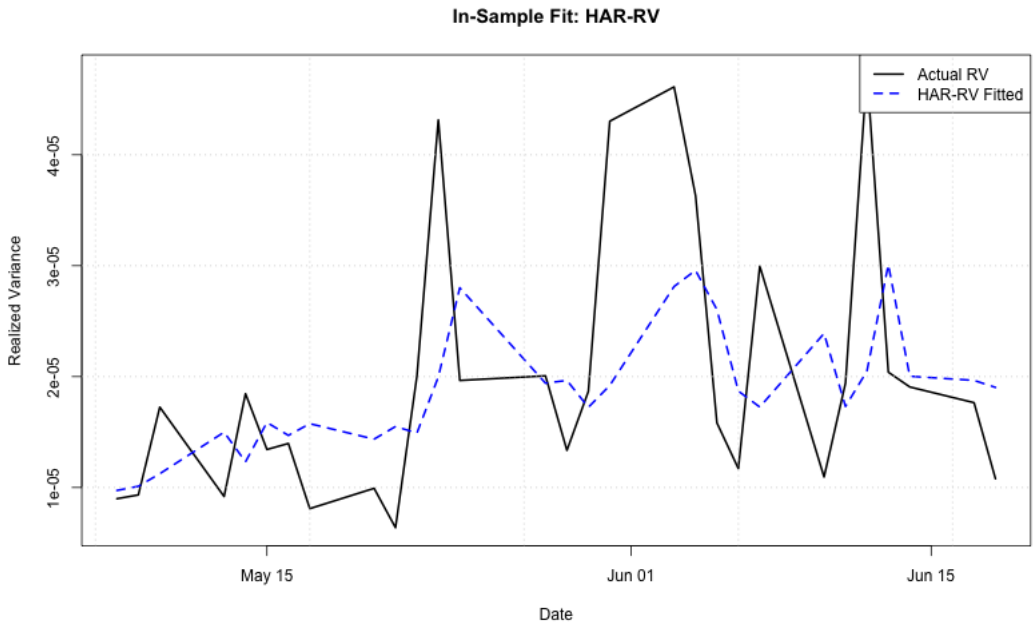


Figure 5: In-Sample Fit: Actual RV vs HAR-RV Fitted Values

5.2 In-Sample Estimation

Tables 3 and 4 show the in-sample estimation results for the three models. The GARCH(1,1) yields the standard result of strong volatility persistence, with $\hat{\alpha}_1 = 0.044$ and $\hat{\beta}_1 = 0.935$.

However, α_1 is not statistically significant, which is likely due to the small sample size (66 daily observations).

Variable	Est_HAR	Pval_HAR	Est_HARX	Pval_HARX
Intercept	1.85e-05	0.026	1.88e-05	0.154
$RV_{t-1}^{(d)}$	3.66e-01	0.027	3.42e-01	0.181
$RV_{t-1}^{(w)}$	-5.46e-02	0.823	-4.30e-02	0.865
$RV_{t-1}^{(m)}$	-2.05e-01	0.295	-2.21e-01	0.392
Vol_{t-1}	–	–	1.40e-13	0.731
Txn_{t-1}	–	–	-1.36e-11	0.853
Adj. R^2	0.115	–	0.072	–
F p-value	–	0.048	–	0.166

Table 3: In-Sample Estimation Results: HAR-RV and HAR-RV-X

For the HAR-RV model, only the daily lag $RV_{t-1}^{(d)}$ is statistically significant ($\hat{\beta}_1 = 0.366$, $p = 0.027$), while the weekly and monthly components are not. The model explains 17.7% of the variance in daily RV (adjusted $R^2 = 0.115$), and the overall F-test is significant ($p = 0.048$). The weak role of the weekly and monthly components is consistent with the limited autocorrelation structure we observed in Section 4. Adding volume and transactions in the HAR-RV-X model does not improve the fit. Neither activity variable is significant, and the adjusted R^2 drops to 0.072. Moreover, the overall F-test is no longer significant ($p = 0.166$).

	Estimate	Std. Error	p-value
μ	0.1720	0.0690	0.0127
ω	0.0000	0.0008	1.0000
α_1	0.0442	0.0440	0.3150
β_1	0.9353	0.0436	0.0000

Table 4: In-Sample Estimation Results: GARCH(1,1)

5.3 Out-of-Sample Forecasting

Table 5 reports the out-of-sample forecast evaluation over the last 15 trading days of the sample. Contrary to hypothesis H1, the GARCH(1,1) outperforms the HAR-RV model across all three metrics, with a lower RMSE (9.14×10^{-6} vs 1.09×10^{-5}), lower MAE (7.13×10^{-6} vs 9.29×10^{-6}), and a positive out-of-sample R^2 of 0.220, while both HAR models get negative R^2 values. Our second hypothesis (H2) is also rejected, as the HAR-RV-X performs marginally worse than the HAR-RV. Hence, in our setting, volume and transactions add no predictive value.

Table 5: Out-of-Sample Forecast Evaluation (15 days)

Model	RMSE	MAE	R^2_{OOS}
GARCH(1,1)	9.14×10^{-6}	7.13×10^{-6}	0.220
HAR-RV	1.09×10^{-5}	9.29×10^{-6}	-0.104
HAR-RV-X	1.10×10^{-5}	9.39×10^{-6}	-0.132

We see on Figure 6 that the HAR-RV and HAR-RV-X forecasts are nearly identical and remain relatively flat, failing to track the movements in actual RV. The GARCH(1,1) forecast also stays smooth but captures the declining trend of actual RV over the out-of-sample period. Overall, all the three models fail to capture the RV variations. Those failures are likely explain by the small sample size, a point we further discuss later.

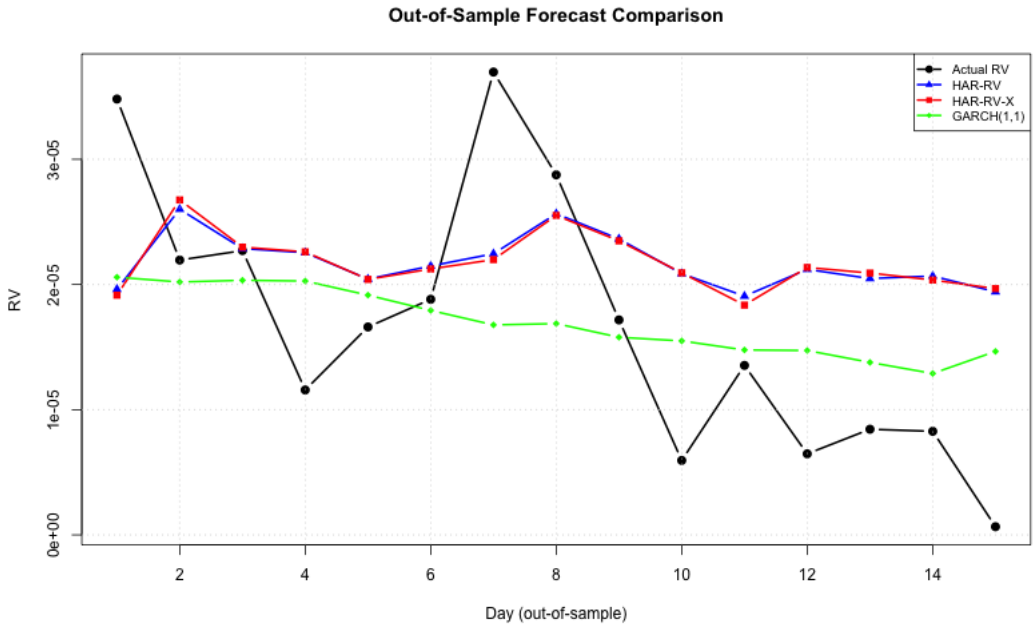


Figure 6: Out-of-Sample Forecast Comparison

The Diebold-Mariano tests confirm that none of the forecast differences are statistically significant at conventional levels (all p-values > 0.05). This means, we cannot formally reject equal predictive accuracy between any pair of models.

Table 6: Diebold-Mariano Test Results

Comparison	DM Statistic	p-value
HAR-RV vs GARCH(1,1)	1.351	0.177
HAR-RV-X vs GARCH(1,1)	1.482	0.138
HAR-RV-X vs HAR-RV	1.709	0.087

5.4 Robustness Check: Intraday HAR Model

To address the small sample limitation of the daily HAR model, we specify an intraday version of the HAR specification using 30-minute realized variance computed from 5-minute returns. This yields approximately 650 in-sample observations compared to 29 in the daily version, where 22 days are lost to the specification of the monthly HAR component and 15 are reserved for out-of-sample evaluation, all while preserving the 5-minute sampling frequency we chose and avoiding microstructure noise. We redefined the three HAR components as follows: the short component corresponds to 30-minutes, the medium component to half a day, and the long component to full trading day.

The model becomes:

$$RV_t = \beta_0 + \beta_1 RV_{t-1}^{(30m)} + \beta_2 RV_{t-1}^{(hd)} + \beta_3 RV_{t-1}^{(d)} + \varepsilon_t \tag{5}$$

where $RV_{t-1}^{(30m)}$ is the previous 30-minute block, $RV_{t-1}^{(hd)}$ is the half-day average, and $RV_{t-1}^{(d)}$ is the full-day average. This is similar to the original HAR model of Corsi (2009), but we modified the horizons.

Table 7: Robustness: Intraday HAR-RV (30-minute blocks)

Component	Estimate	p-value
Intercept	7.52×10^{-7}	0.001
Short (30-min)	2.18×10^{-1}	< 0.001
Medium (half-day)	1.58×10^{-1}	0.133
Long (full day)	3.32×10^{-1}	0.005
Adj. R^2	0.177	
F p-value	< 0.001	
R^2_{OOS}	0.111	

Strikingly, the results are stronger than the daily specification. The F-test is highly significant ($p < 0.001$), and both the short and long components are statistically significant. The out-of-sample R^2 is positive at 0.111, meaning the model outperforms a naive forecast based on the historical mean. These results suggest that the weak performance of the daily HAR model is primarily attributable to the small sample size rather than a failure of the HAR framework itself.

5.5 Robustness Check: HAR Model Without Monthly Component

A second robustness check is motivated by two observations. First, as discussed in Section 4.3, the ACF of daily RV shows significant autocorrelations only up to approximately 5 lags, providing no empirical support for the monthly component in our sample. Second,

the monthly component requires 22 days of initialization, reducing the in-sample period to only 29 observations. We therefore estimate a restricted HAR model using only the daily and weekly components:

$$RV_t = \beta_0 + \beta_1 RV_{t-1}^{(d)} + \beta_2 RV_{t-1}^{(w)} + \varepsilon_t \tag{6}$$

This recovers 17 additional observations, bringing the in-sample period to 46 days.

Table 8: Robustness: HAR-DW vs GARCH(1,1) Out-of-Sample

Model	Adj. R^2 (in-sample)	RMSE	R^2_{OOS}
GARCH(1,1)	–	9.14×10^{-6}	0.220
HAR-RV (full)	0.115	1.09×10^{-5}	-0.104
HAR-DW	0.462	1.02×10^{-5}	0.026

The results improve substantially. The in-sample adjusted R^2 rises from 0.115 to 0.462, the daily component becomes highly significant ($p < 0.001$), and the out-of-sample R^2 turns positive at 0.026. While GARCH still outperforms out-of-sample with an R^2_{OOS} of 0.220, the gap narrows considerably compared to the full HAR model. This confirms that the monthly component was harmful in our setting, both because it lacked empirical support in the autocorrelation structure and because it consumed observations that a small sample cannot afford to lose.

6 Conclusion

This paper investigates whether high-frequency data and market activity variables can improve S&P500 volatility forecasting relative to a classical GARCH(1,1) benchmark. Our results do not support either of our hypothesis. H1 is rejected: the HAR-RV model does not outperform GARCH(1,1) out-of-sample. The GARCH produces lower forecast errors and a positive out-of-sample R^2 of 0.220, while both HAR models yield negative values, meaning they perform worse than a naive forecast based on the historical mean. H2 is also rejected: augmenting the HAR-RV with volume and transaction counts yields no improvement in forecasting accuracy.

Nonetheless, these results should be interpreted with caution. Our sample covers only 66 trading days. Three pieces of evidence suggest that sample size, rather than model misspecification, is responsible for the weak HAR performance. First, with only 29 in-sample observations, the monthly component of the HAR model has no empirical support, as confirmed by the ACF analysis in Section 4.3. Second, after removing the monthly component and estimating HAR model with only the daily and weekly components on 46 observations, it raises the in-sample adjusted R^2 from 0.115 to 0.462 and produces a posi-

tive out-of-sample R^2 of 0.026. Although it is still small, this substantially reduces the gap with the GARCH. Third, when the HAR framework is applied at the intraday level using 30-minutes blocks and 650 observations, both the short and long components become highly significant and the out-of-sample R^2 reaches 0.111.

Finally, although results have failed, the pipeline developed in this project can be fully reusable. Therefore, we recommend reproducing the analysis to a longer sample of several years, which would allow for a more rigorous evaluation of all three models. Furthermore, several extensions could also enrich the analysis. For instance, one could add jump components in the model to model the largest spikes. Finally, machine learning methods such as LASSO or random forests could be applied to a richer set of activity variables to determine which predictors are most relevant.

7 AI Acknowledgment

- Claude has been used to help with the data mining part using the API.
- Claude has been used for coding the robustness checks:
  - "Given our code, write a code chunk that does the HAR-RV model without the monthly component (use only daily and weekly ones)."
  - "Given our code, write a code chunk that does the HAR-RV model but instead of using daily, weekly and monthly component, use the 30 minutes, half a day and a full day."
  - Claude has been used to gain time in displaying results in a more readable way:
    * "Display the results after each chunk of codes using cat() function."
    * "For the models results, build and save the tables in the most readable way, such that we can include them with `\input{}` in our overleaf document, and that old tables are overwritten if the code is re-run."

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